FEU Institute of Technology
COLLEGE OF ENGINEERING • COLLEGE OF COMPUTER STUDIES

Computer System and Architecture

Machine Problem

FLAB3

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Data Performed	Date Submitted

List the steps that you perform to come up with the program.

Step 1: Architectural Alignment & Environment Setup

- Target Environment: Configured the program for a 64-bit x86_64 Linux environment (_start entry point, syscall interface) to match OneCompiler's default execution model.
- Memory Organization: Divided memory into a .data section for static strings, banners, and the per-unit rate (2108), and a .bss section (resb) for dynamic user inputs (input_buffer, student details, and calculation buffers).

Step 2: Implementing the MACRO (print_string)

- Designed a reusable macro %macro print_string 1 to handle string output cleanly.
- Inside the macro, implemented a dynamic null-terminator search loop (%%calc_len) to calculate string lengths on the fly, followed by a 64-bit Linux sys_write system call (mov rax, 1, mov rdi, 1) while safely preserving registers via push/pop.

Step 3: Streamlined Multiline Input Reading

- Instead of multiple blocking sys_read calls (which cause timeouts in web-based sandboxes like OneCompiler), implemented a single, unified input read (mov rax, 0) into a 512-byte input_buffer.
- Created the parse_inputs procedure and helper extract_line routine to parse the multiline STDIN buffer sequentially line-by-line (handling student number, first name, middle name, last name, program, year level, and units).

Step 4: Implementing the ASCII-to-Integer Procedure (ascii_to_int)

- Built a procedure to convert the ASCII string representation of units (e.g., "18") into a raw numeric integer in RAX.
- Iterated through the character buffer, validated numeric digits between '0' and '9', and performed standard base-10 accumulation (imul rax, rax, 10 + add rax, rcx).

Step 5: Implementing the Tuition Calculation Procedure (compute_tuition)

- Designed a dedicated procedure to multiply the integer unit count by the fixed rate per unit (2108).
- Implemented an integer-to-ASCII conversion routine using repeated division by 10 (div rcx), pushing remainders onto the stack/buffer to format the calculated total (e.g., 37944) back into a printable string (tuition_result).

Step 6: COR Summary Layout & Formatting

- Structured the final output block to mimic the official portal profile and assessment layout (incorporating enrollment term status, student credentials, curriculum code, and the formatted tuition fee).

PNG+ 1

- Used precise label arrangements combined with comma/space separators and newlines (10) for clean terminal readability.

Step 7: Debugging and Syntax Refining

- Addressed NASM syntax strictness by removing stray trailing commas in data definitions (e.g., on lbl_term and lbl_status) that caused parser instruction errors.

- Ensured clean stack framing (push rbp, mov rbp, rsp) and proper register preservation across all procedures and macros.

Paste the source code and explain the functionalities and take a screenshot of the final output.

```
;
==========================================================================
; FEU TECH - Certificate of Registration (COR) & Tuition Calculator
;
==========================================================================

section .data
   banner        db  "=========================================", 10
                 db  "    FEU INSTITUTE OF TECHNOLOGY        ", 10
                 db  "    CERTIFICATE OF REGISTRATION (COR)   ", 10
                 db  "=========================================", 10, 0

   lbl_header     db  10, "== COR SUMMARY / USER PROFILE ==", 10, 0
   lbl_term       db  "Enrollment Status: Term 1 SY 2026-2027", 10, 0
   lbl_stud_no    db  "Student Number   : ", 0
   lbl_name       db  "Student Name     : ", 0
   lbl_program    db  "Program          : ", 0
   lbl_curric     db  "Curriculum Code  : BSCSSE2025", 10, 0
   lbl_year       db  "Year Level       : ", 0
   lbl_status     db  "Student Type     : REGULAR", 10, 0
   lbl_units      db  "Enrolled Units   : ", 0
   lbl_tuition    db  "Tuition Fee      : PHP ", 0
   comma_space    db  ", ", 0
   space          db  " ", 0
   newline        db  10, 0
   rate_per_unit  dq  2108       ; Rate per unit

section .bss
   input_buffer   resb 512
   stud_id        resb 32
   first_name     resb 32
   middle_name    resb 32
   last_name      resb 32
   program        resb 16
```

```
    year_level      resb 8
    input_units     resb 8
    tuition_result  resb 16

%macro print_string 1
    push rax
    push rbx
    push rcx
    push rdx
    push rsi
    push rdi

    mov rsi, %1
    xor rdx, rdx
%%calc_len:
    cmp byte [rsi + rdx], 0
    je %%do_print
    inc rdx
    jmp %%calc_len

%%do_print:
    mov rax, 1          ; sys_write
    mov rdi, 1          ; stdout
    syscall

    pop rdi
    pop rsi
    pop rdx
    pop rcx
    pop rbx
    pop rax
%endmacro

section .text
    global _start

_start:
    print_string banner

    mov rax, 0          ; sys_read
    mov rdi, 0          ; stdin
    mov rsi, input_buffer
    mov rdx, 511
```

```
syscall

call parse_inputs

mov rsi, input_units
call ascii_to_int

mov rdi, rax
call compute_tuition

print_string lbl_header
print_string lbl_term

print_string lbl_stud_no
print_string stud_id
print_string newline

print_string lbl_name
print_string last_name
print_string comma_space
print_string first_name
print_string space
print_string middle_name
print_string newline

print_string lbl_program
print_string program
print_string newline

print_string lbl_curric
print_string lbl_year
print_string year_level
print_string newline

print_string lbl_status
print_string lbl_units
print_string input_units
print_string newline

print_string lbl_tuition
print_string tuition_result
print_string newline
```

```
    ; Exit Program (sys_exit)
    mov rax, 60
    xor rdi, rdi
    syscall

parse_inputs:
    mov rsi, input_buffer

    mov rdi, stud_id
    call extract_line

    mov rdi, first_name
    call extract_line

    mov rdi, middle_name
    call extract_line

    mov rdi, last_name
    call extract_line

    mov rdi, program
    call extract_line

    mov rdi, year_level
    call extract_line

    mov rdi, input_units
    call extract_line
    ret

extract_line:
.loop:
    mov al, [rsi]
    cmp al, 10
    je .done_line
    cmp al, 13
    je .done_line
    cmp al, 0
    je .done_all
    mov [rdi], al
    inc rsi
    inc rdi
    jmp .loop
```

```
.done_line:
    mov byte [rdi], 0
    inc rsi
    cmp byte [rsi], 10
    jne .skip_lf
    inc rsi
.skip_lf:
    ret
.done_all:
    mov byte [rdi], 0
    ret

compute_tuition:
    push rbp
    mov rbp, rsp
    push rax
    push rbx
    push rcx
    push rdx
    push rsi
    push rdi

    mov rax, rdi
    mov rbx, [rate_per_unit]
    mul rbx                 ; RAX = units * rate_per_unit

    mov rdi, tuition_result + 15
    mov byte [rdi], 0
    dec rdi
    mov rcx, 10

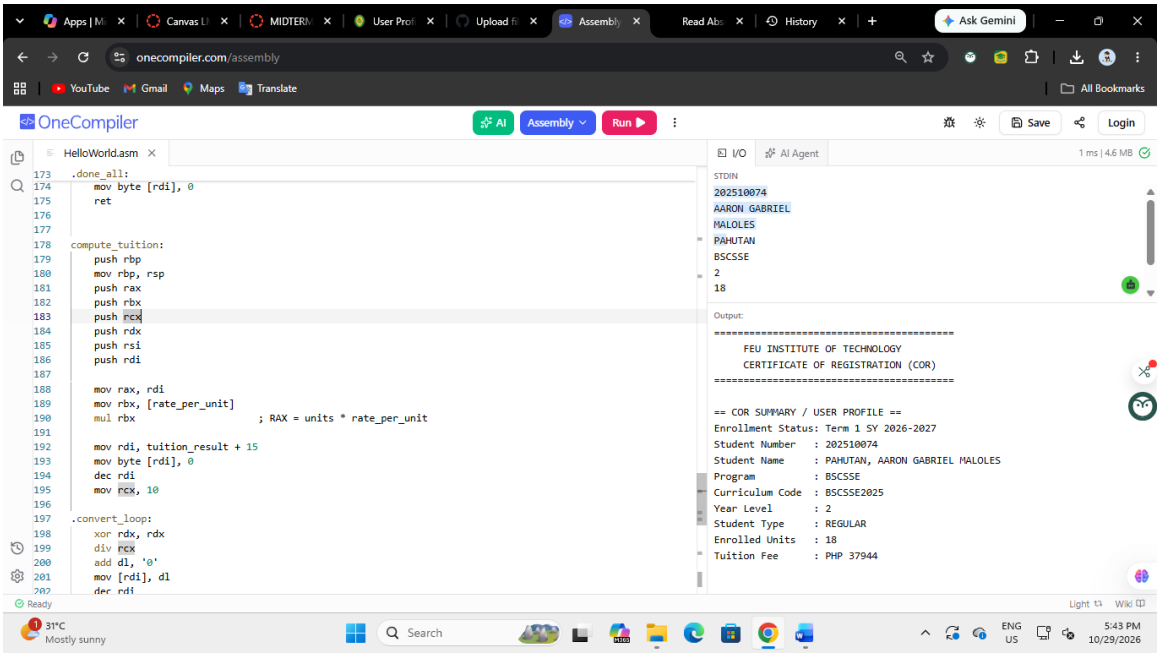
.convert_loop:
    xor rdx, rdx
    div rcx
    add dl, '0'
    mov [rdi], dl
    dec rdi
    test rax, rax
    jnz .convert_loop

    inc rdi
    mov rsi, rdi
    mov rdi, tuition_result
```

```
.copy_res:
    lodsb
    stosb
    test al, al
    jnz .copy_res

    pop rdi
    pop rsi
    pop rdx
    pop rcx
    pop rbx
    pop rax
    mov rsp, rbp
    pop rbp
    ret

ascii_to_int:
    xor rax, rax
.next_digit:
    movzx rcx, byte [rsi]
    cmp rcx, '0'
    jl .done_conv
    cmp rcx, '9'
    jg .done_conv
    sub rcx, '0'
    imul rax, rax, 10
    add rax, rcx
    inc rsi
    jmp .next_digit
.done_conv:
    ret
```


onecompiler.com/assembly
OneCompiler
AI
Assembly
Run
Save
Login
HelloWorld.asm
.done_all:
mov byte [rdi], 0
ret
compute_tuition:
push rbp
mov rbp, rsp
push rax
push rbx
push rcx
push rdx
push rsi
push rdi
mov rax, rdi
mov rbx, [rate_per_unit]
mul rbx ; RAX = units * rate_per_unit
mov rdi, tuition_result + 15
mov byte [rdi], 0
dec rdi
mov rcx, 10
.convert_loop:
xor rdx, rdx
div rcx
add dl, '0'
mov [rdi], dl
dec rdi
I/O
AI Agent
STDIN
202510074
AARON GABRIEL
MALOLES
PAHUTAN
BSCSSE
2
18
Output:
FEU INSTITUTE OF TECHNOLOGY
CERTIFICATE OF REGISTRATION (COR)
== COR SUMMARY / USER PROFILE ==
Enrollment Status: Term 1 SY 2026-2027
Student Number : 202510074
Student Name : PAHUTAN, AARON GABRIEL MALOLES
Program : BSCSSE
Curriculum Code : BSCSSE2025
Year Level : 2
Student Type : REGULAR
Enrolled Units : 18
Tuition Fee : PHP 37944
Ready
31°C
Mostly sunny
Search
ENG
US
5:43 PM
10/29/2026